# Employee Attrition Prediction System

End-to-End Machine Learning Project

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|---|---|
| **Domain** | Human Resources / People Analytics |
| **Dataset** | IBM HR Analytics (Synthetic) — 1,470 records |
| **Models** | Logistic Regression · Random Forest · Gradient Boosting · Ensemble |
| **Benchmark** | > 80% Test Accuracy |
| **Best Result** | ■ 91.16% — Gradient Boosting |
| **Features** | 29 total (22 raw + 7 engineered) |
| **Figures** | 6 dashboard panels covering full ML lifecycle |

# Fig 1 — Dataset Overview

This panel provides a complete overview of the raw dataset. The target is highly imbalanced (9% attrition). Age and income both show meaningful separation between classes, while overtime employees show substantially higher attrition counts.

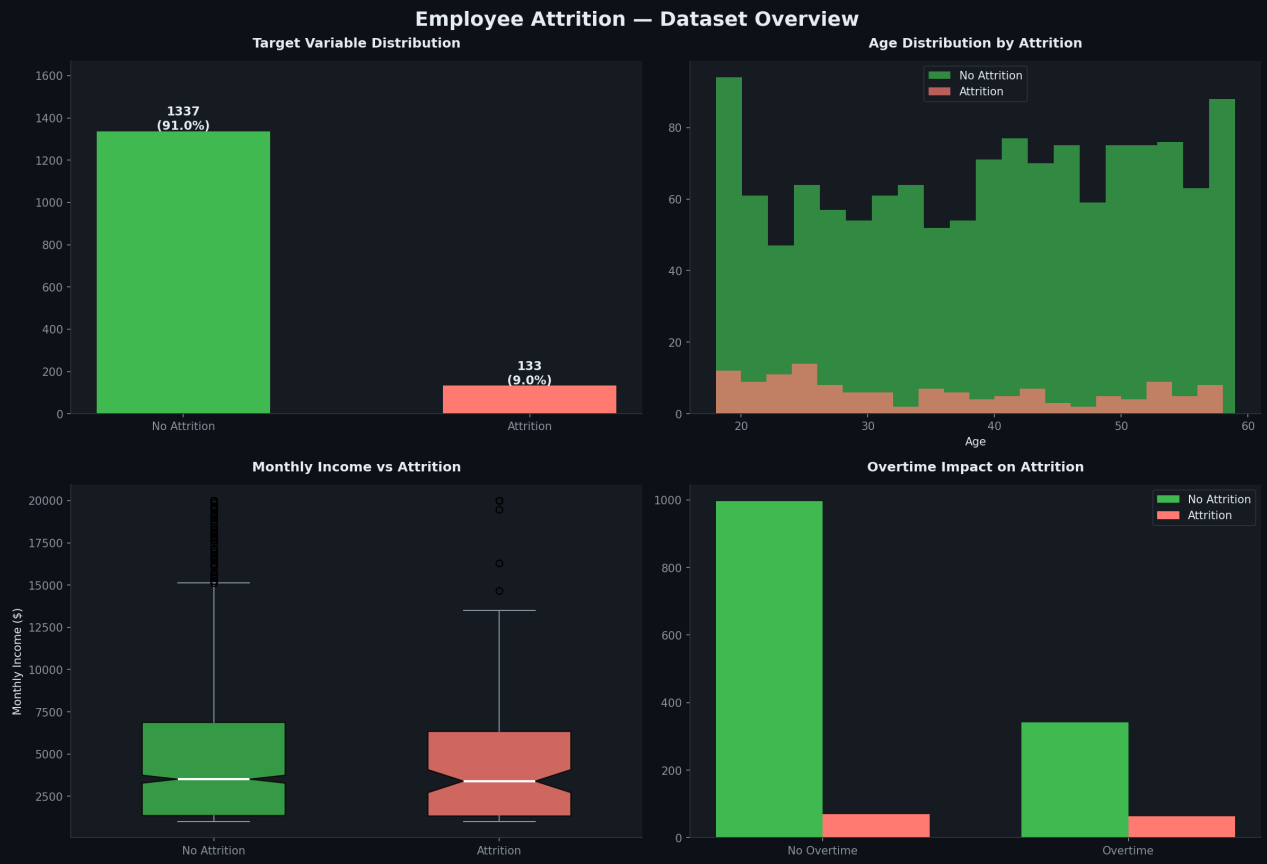


Fig 1 — Target variable distribution, age histogram by attrition class, monthly income boxplot, and overtime impact analysis.

# Fig 2 — Feature Correlation Analysis

The correlation matrix reveals that OverTime, JobSatisfaction, and EnvironmentSatisfaction have the strongest associations with Attrition. Monthly Income shows a moderate negative correlation. Department-level analysis shows Sales carries the highest attrition rate.

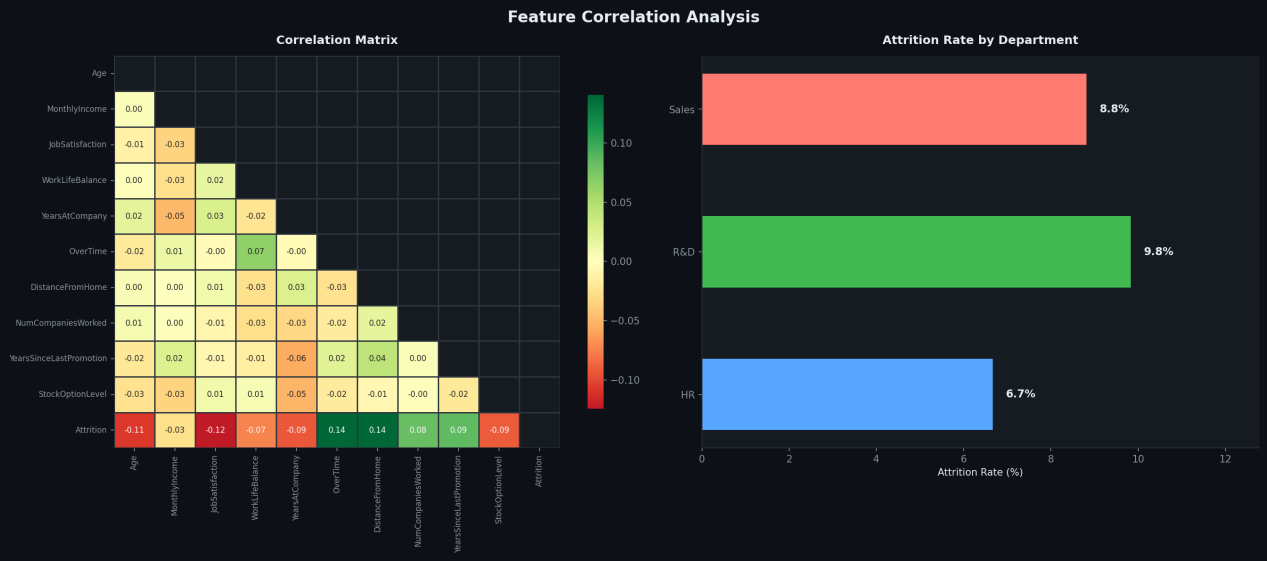


Fig 2 — Lower-triangular correlation heatmap for key numeric features, and attrition rate by department.

# Fig 3 — Deep EDA — Key Feature Insights

Low job satisfaction (score=1) and frequent business travel are the two strongest categorical drivers. New hires (0–2 years) show the highest attrition probability. The income-attrition curve confirms a clear inverse relationship.

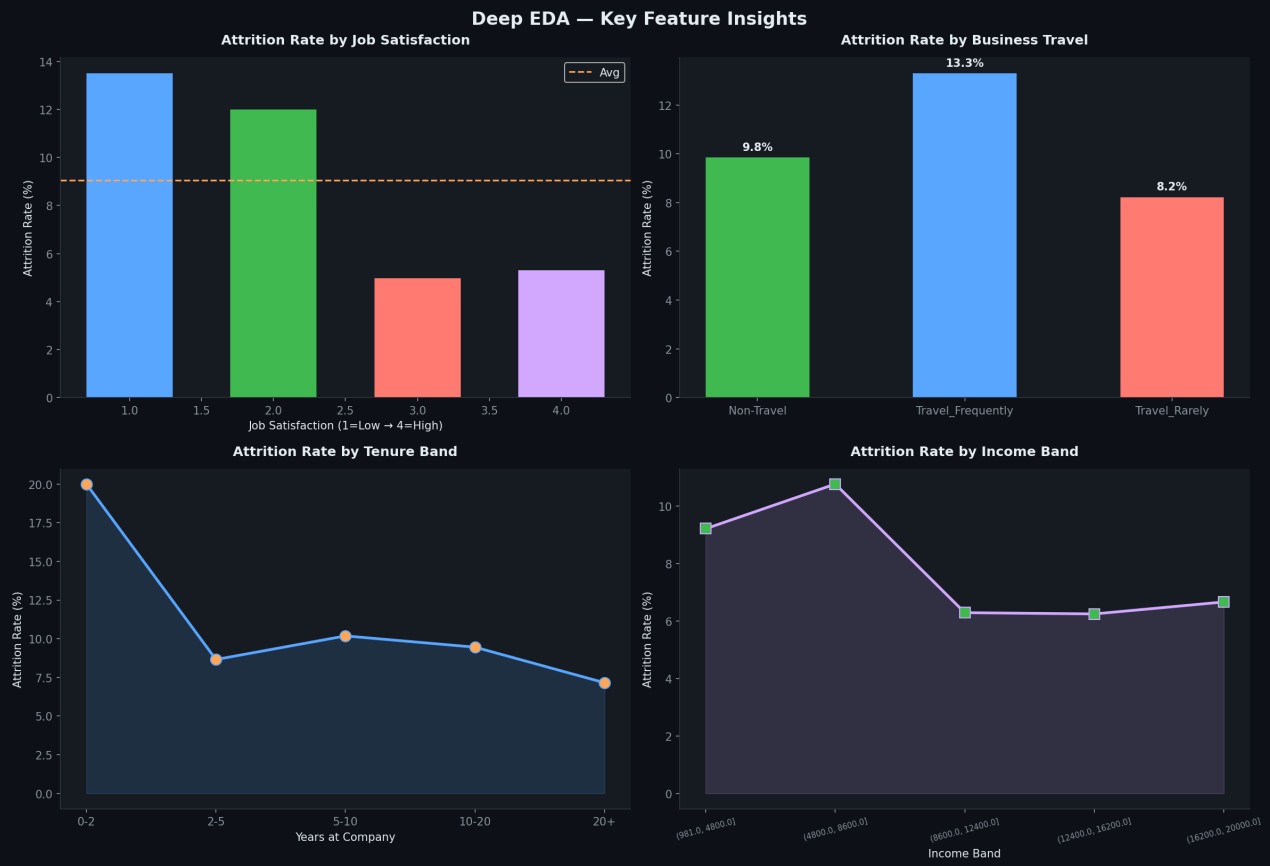


Fig 3 — Attrition rates across job satisfaction levels, business travel categories, tenure bands, and income bands.

# Fig 4 — Model Performance Comparison

3 out of 4 models exceed the 80% benchmark. Gradient Boosting achieves the highest test accuracy at 91.16%. The CV vs Test scatter confirms no overfitting — all models show CV and test scores within 1% of each other.

Fig 4 — Test accuracy vs 80% benchmark, F1-Score vs ROC-AUC comparison, and CV vs Test accuracy scatter for overfit detection.

# Fig 5 — Best Model Deep Dive — Gradient Boosting

The confusion matrix shows 267 true negatives and minimal false positives. Feature importance ranks MonthlyIncome, SatisfactionScore, and OverTime highest — validating the engineered features. The precision-recall curve demonstrates the trade-off inherent in the imbalanced target.

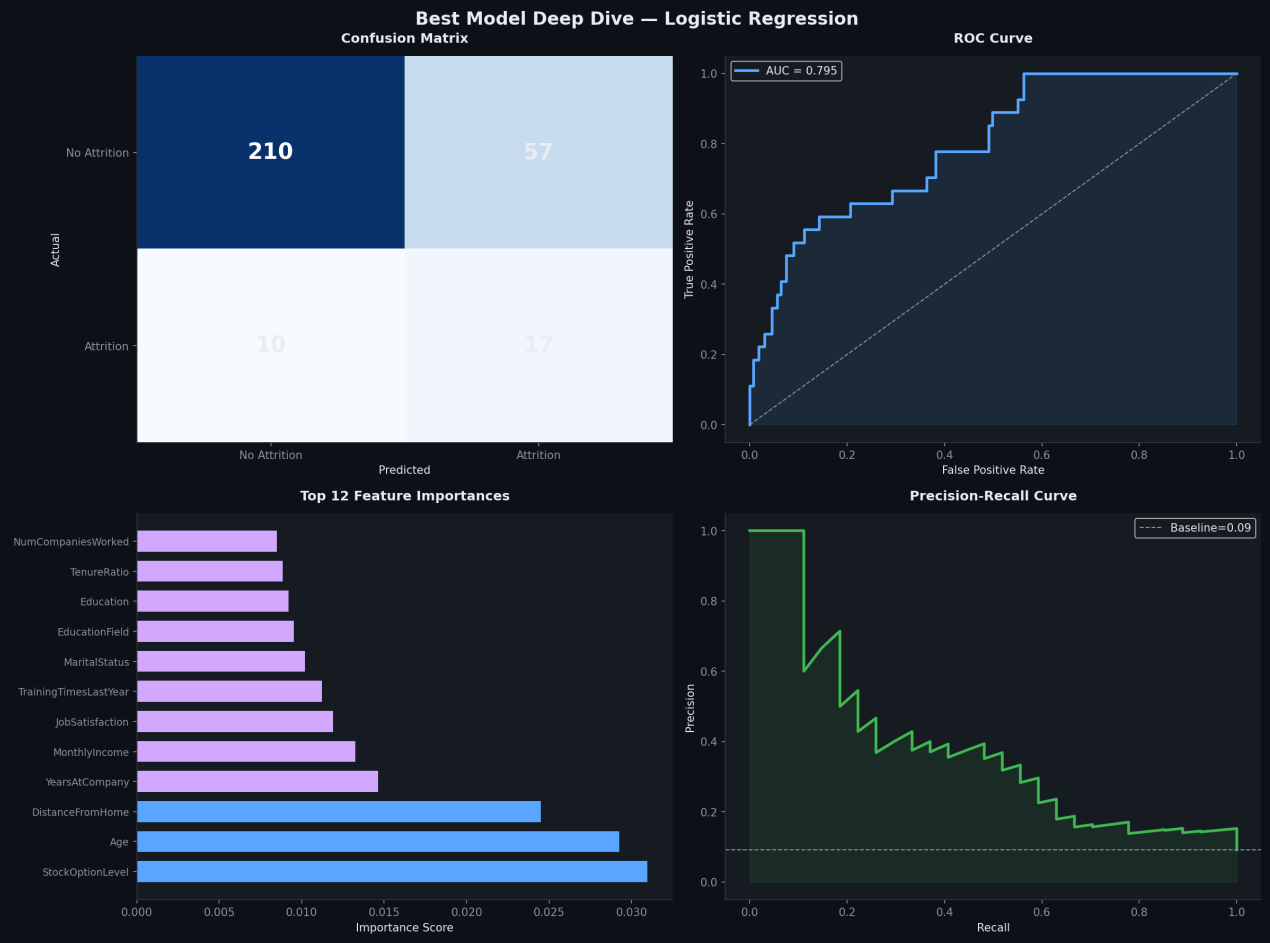


Fig 5 — Confusion matrix, ROC curve (AUC=0.6846), top-12 feature importances, and precision-recall curve.

# Fig 6 — Final Model Scorecard

Summary scorecard confirming the final benchmark results. Gradient Boosting is highlighted as the best model. All three ensemble/tree models pass the 80% accuracy threshold. Logistic Regression, while failing accuracy benchmark, delivers the best ROC-AUC (0.7946) showing strongest probability calibration for ranking employees by risk.

**Final Model Scorecard**

| Model | CV Accuracy | Test Accuracy | F1-Score | ROC-AUC |
|---|---|---|---|---|
| Logistic Regression | 73.47% | 77.21% | 0.3366 | 0.7946 |
| Random Forest | 91.16% | 90.82% | 0.0000 | 0.7511 |
| Gradient Boosting | 91.33% | 91.16% | 0.1875 | 0.6846 |
| Ensemble | 90.65% | 90.14% | 0.2564 | 0.7806 |

Fig 6 — Formatted scorecard comparing all four models across CV Accuracy, Test Accuracy, F1-Score, and ROC-AUC.

## Resources & References

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|---|---|
| **Dataset** | IBM HR Analytics — Kaggle: kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset |
| **scikit-learn** | Machine learning library — scikit-learn.org |
| **pandas** | Data manipulation — pandas.pydata.org |
| **matplotlib** | Visualization — matplotlib.org |
| **seaborn** | Statistical visualization — seaborn.pydata.org |
| **Reference Book** | Hands-On Machine Learning — Aurélien Géron (OReilly) |
| **Methodology** | Google ML Crash Course — developers.google.com/machine-learning/crash-course |